

## Machine Learning Home Work

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### Problem 1.2:

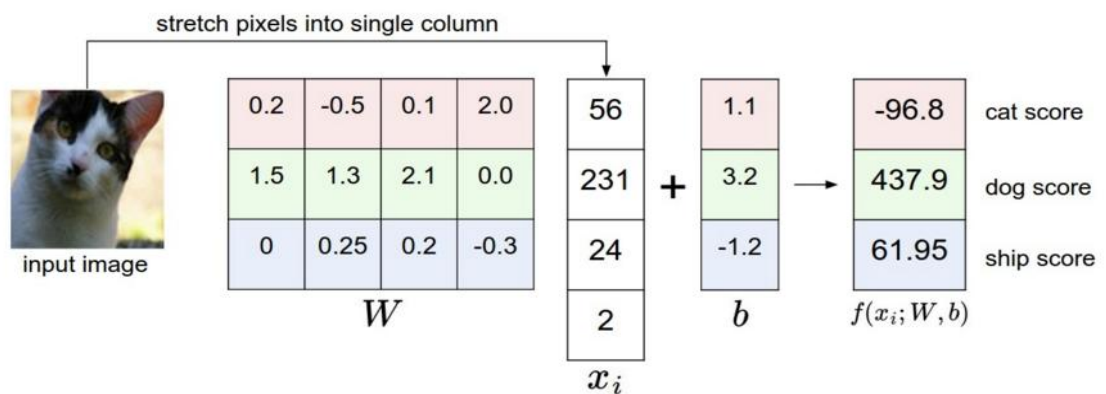
The task is, an image classification model takes a single test image and classifies either it is from the given class or not by linear regression classifier; if image is classified into “correctly”, then label it as “1”, otherwise, label “0”.

**Task T:** Classify the images from CIFAR-10 data base

**Performance P:** Percentage of images that correctly Classify by classifier

**Experiment E:** Classification on training data

$$f(x_i, W, b) = Wx_i + b \quad : \text{Linear classifier}$$



Mapping an image to class scores

Tradeoffs considered in formulating this learning task.

- **Viewpoint variation.** A single instance of an object can be oriented in many ways with respect to the camera.
- **Scale variation.** Visual classes often exhibit variation in their size (size in the real world, not only in terms of their extent in the image).
- **Deformation.** Many objects of interest are not rigid bodies and can be deformed in extreme ways.
- **Occlusion.** The objects of interest can be occluded. Sometimes only a small portion of an object (as little as few pixels) could be visible.
- **Illumination conditions.** The effects of illumination are drastic on the pixel level.
- **Background clutter.** The objects of interest may *blend* into their environment, making them hard to identify.
- **Intra-class variation.** The classes of interest can often be relatively broad, such as *chair*. There are many different types of these objects, each with their own appearance.
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**Problem 1.3:**

Rule for estimating training values:  $V_{train} \leftarrow \hat{v}(\text{Successor}(b))$

And we have  $\hat{V}(b) = w_0 + w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + w_5x_5 + w_6x_6$

Square error:  $E = \sum_{\langle b, V_{train}(b) \rangle \in \text{trainingexample}} (V_{train}(b) - \hat{V}(b))^2$

**Target:** Minimize the Cost function

Differentiate the error w.r.t. weight calculate Gradient Decent

And set learning rate to reach global minima,

$$\frac{\partial E}{\partial w_i} = 2(V_{train}(b) - \hat{V}(b)) \frac{\partial (V_{train}(b) - \hat{V}(b))}{\partial w_i} = \begin{cases} 2(V_{train}(b) - \hat{V}(b))(-X_i) & (i \neq 0) \\ -2(V_{train}(b) - \hat{V}(b)) & (i = 0) \end{cases}$$

LMS shows:  $w_i \leftarrow w_i + \eta(V_{train}(b) - \hat{V}(b))X_i$

Then we get  $w_i \leftarrow w_i + \left(\frac{1}{2}\eta\right) \frac{-\partial E}{\partial w_i}$

LMS training rule alters weights in this proportion for each training example it encounters.